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University of Plymouth

MODULE CODE:	**ELEC345 Examination**
TITLE OF PAPER:	**High Speed Communications**

TIME ALLOWED	**Three hours**
DATE	**Thursday 18 January 2024**
TIME	**09:00 – 12:00**
FACULTY	**Science and Engineering**
SCHOOL	**Engineering, Computing and Mathematics**
ACADEMIC YEAR	**2023/24**
STAGE	**Three**

INSTRUCTIONS TO CANDIDATES:

Candidates should attempt **FOUR** out of the **FIVE** questions.
All questions carry equal marks. Marks for parts of questions are shown where appropriate.
Data provided:
Filter Tables
Equation list
Four full size Smith Charts – please attach to answer book.

Candidates are not permitted to look at the examination paper until instructed to do so.

1. This question is on mobile radio channel properties and diversity techniques.

(a) In a mobile radio channel what do the following terms and expressions mean?

i. Fast and slow fading
ii. Multipath
iii. Dopler spread.
iv. Frequency selective fading
v. Flat fading

[10 marks]

(b) List the five diversity techniques used to combat multipath effects.

[5 marks]

(c) Given diversity inputs of 3dB, 5dB 8dB and 10dB give or calculate with an explanation where necessary the outputs of the following diversity combination systems and then how they would change if the 10dB input dropped to 1dB:

i. Switched
ii. Selection
iii. Maximum ratio
iv. Maximum gain

[10 marks]

2. This question is about antenna and stub matching.

(a) Given $\lambda = c/f$ design a half wavelength dipole to operate at a frequency of 1.25 GHz? Assuming the elements are on the end of a piece of rigid coax sketch its design labelling its dimensions.

[5 marks]

(b) When you measure your design using a VNA you find its maximum return loss occurs at 1GHz.

i. Why is this?
ii. Calculate the correct dimensions to meet your original frequency requirement.

[5 marks]

Question 2 continued over…..

Question 2 continued…

(c) You have a patch antenna with an impedance of 100 -100j ohms, calculate both possible parallel single stub matchs for a transmission line with Z_0 = 50 ohm.

[15 marks]

3. This question is about smith charts and matching networks. You may assume a characteristic impedance Z_0 of 50 ohms.

(a) On the Smith charts included with this paper are several labels referring to the scales and areas of the chart. Choose five and explain what each one means/refers to. (A sentence will suffice)

[5 marks]

(b) Plot the load Z_L = 20 - 150j ohms and:

i. Find the angle and magnitude of the voltage reflection coefficient.
ii. The voltage standing wave ratio (VSWR) in linear form and dBs
iii. Plot all values of equal VSWR.
iv. What two points on the VSWR plot correspond to the maximum and minimum voltage.
v. Find/change the load to an admittance (write your answer as well as marking it on the chart)

[10 marks]

(c) For the load in part b (20-150j ohms) design the two possible L matches including the actual component values at a frequency of 200Mhz. Sketch each circuit and label the values.

[10 marks]

4. Channel access and link budgets.

(a) TDMA is a way of providing channel access in cellular systems, give two other methods and explain them along with how the uplink and downlink is organised.

[6 marks]

(b) Sketch a macro cell, micro cell and sectorized antenna system, why are they used?

[4 marks]

(c) What does MIMO stand for and what are its benefits when used in WIFI systems?

[5 marks]

(d) You decide to try to set up a radio link to talk to the international space station. You have a transmitter with a 20W output, a 10dBi directional antenna and a transmit frequency of 1.4GHz with a channel bandwidth of 200kHz. The space station’s antenna is a monopole with a gain of 5dBi and its radio receiver has a gain of 10dB and noise figure of 2dB. It is orbiting at an altitude of 408km.

Calculate:

i. The free space path loss
ii. The received signal strength after the radio receiver P_r
iii. Total noise at the receiver
iv. The signal to noise ratio
v. Give two things you could change to improve the performance.

[10 marks]

5. This question is about filters.

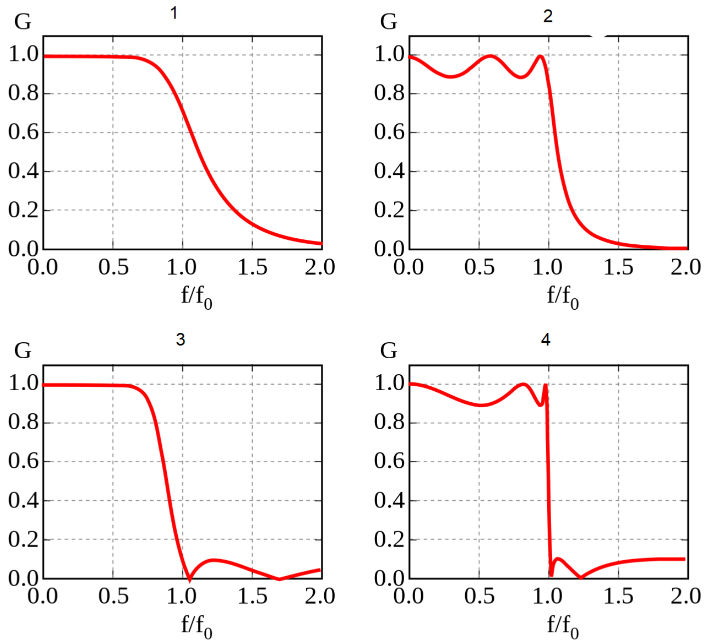


Figure Q5

(a) For the filter bode plots 1-4 shown in Figure Q5 state the filter type that would give this response? Which filter mentioned in lectures is missing?

[5 marks]

(b) What is the benefit of filters 2, 3 and 4 over that of filter 1?

[3 marks]

(c) Using the tables provided design an active low pass Butterworth filter with F_c = 3 MHz and F_s = 9 MHz to achieve A_{min} = 25 dB and 10kΩ scaling resistors.

[7 marks]

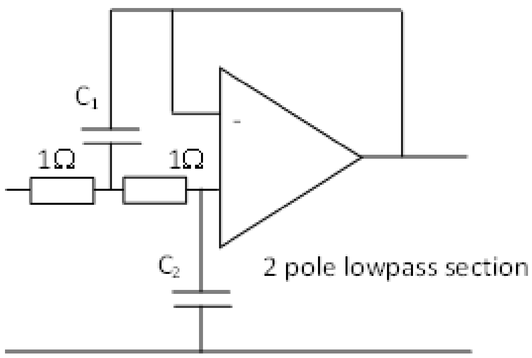
(d) Using the tables provided (on the next page) design an active high pass Butterworth filter with F_s = 1 MHz and F_c = 4 MHz and A_{min} = 55 dB and 10kΩ scaling resistors

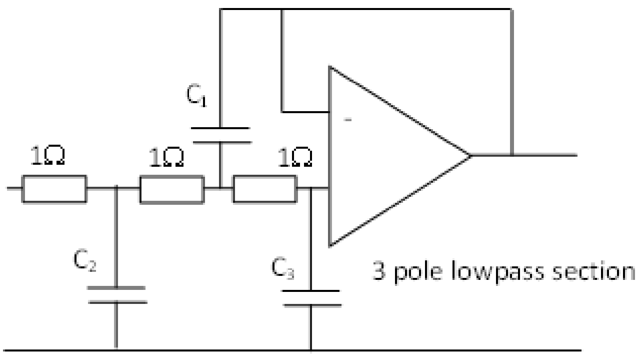
[10 marks]

FILTER DESIGN TABLES

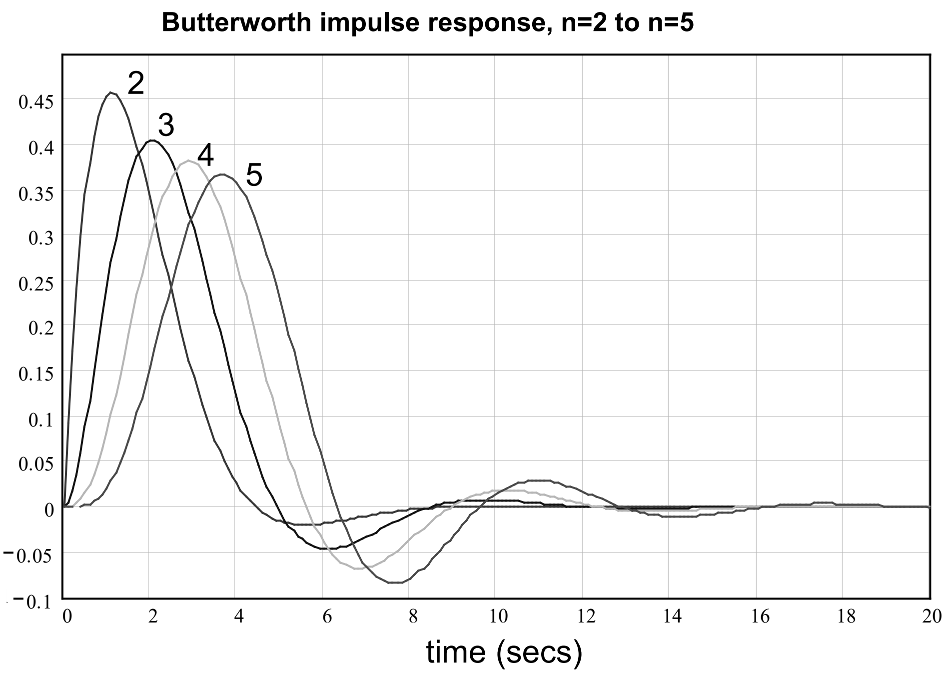
Normalised Butterworth active element values

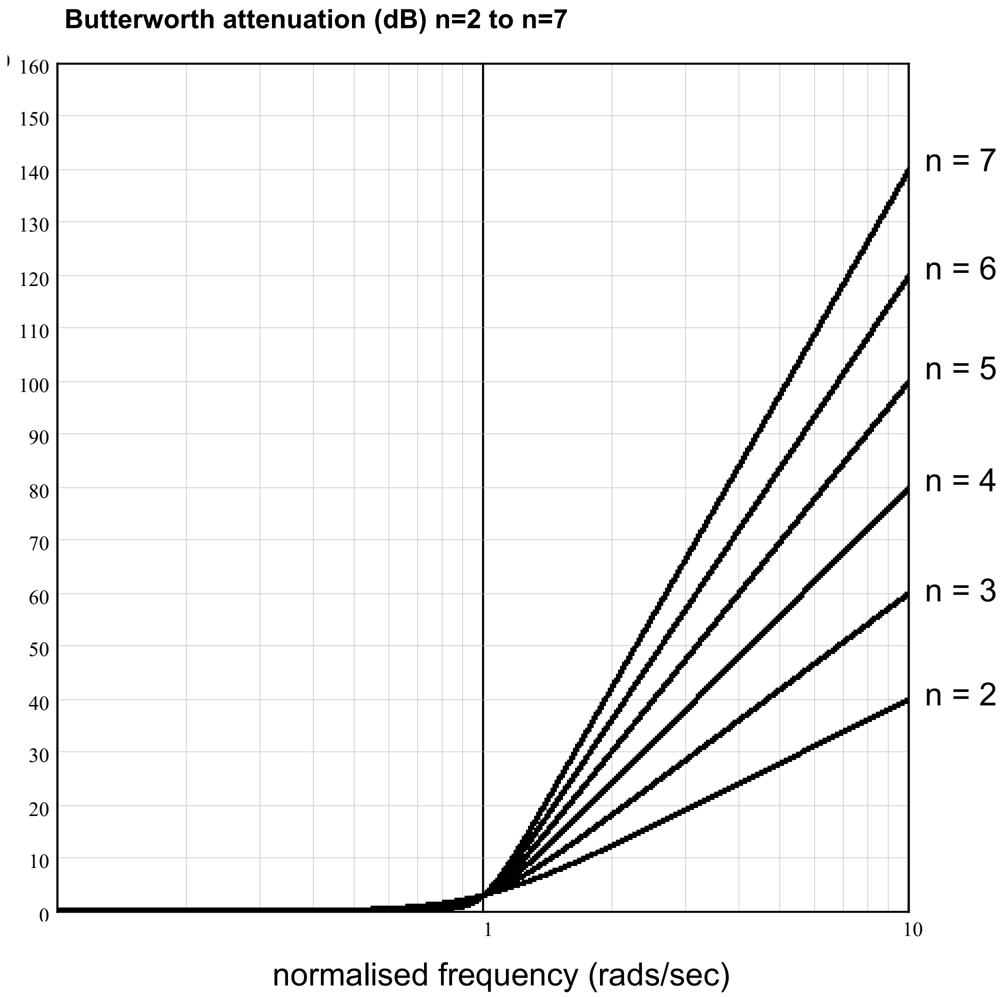
n	C_1	C_2	C_3
2	1.414	0.7071	
3	3.546	1.392	0.2024
4	1.082 2.613	0.9241 0.309	
5	1.753 3.253	1.354 0.309	0.4214
6	1.035 1.414 3.863	0.966 0.7071 0.2588	
7	1.531 1.604 4.493	1.336 0.6235 0.2225	0.4885





Normalised Butterworth impulse response and attenuation characteristics





Useful Equations

$$L \Rightarrow \frac{L.R_o}{2\pi F_c} \qquad C \Rightarrow \frac{C}{2\pi F_c R_o} \qquad R \Rightarrow R.R_o \;\; A_s = \frac{F_s}{F_c}$$

$$R_{HP} = \frac{1}{C_{LP}} \qquad and \qquad C_{HP} = \frac{1}{R_{LP}}$$

$$L = 20\log_{10}\left(\frac{4\pi R}{\lambda}\right)\ dB$$

$\lambda = V_p/f$

$$\gamma_{div,2} = \frac{\gamma_1 + \gamma_2 + 2\sqrt{\gamma_1\gamma_2}}{2} \qquad \gamma_{div,N} = \sum_{i=1}^{N}\gamma_i$$

F = Noise Figure

$$N = -204 + F + 10\log_{10}(B) \quad (dBW)$$

$N = -174 + F + 10\log_{10}(B)$ dBm

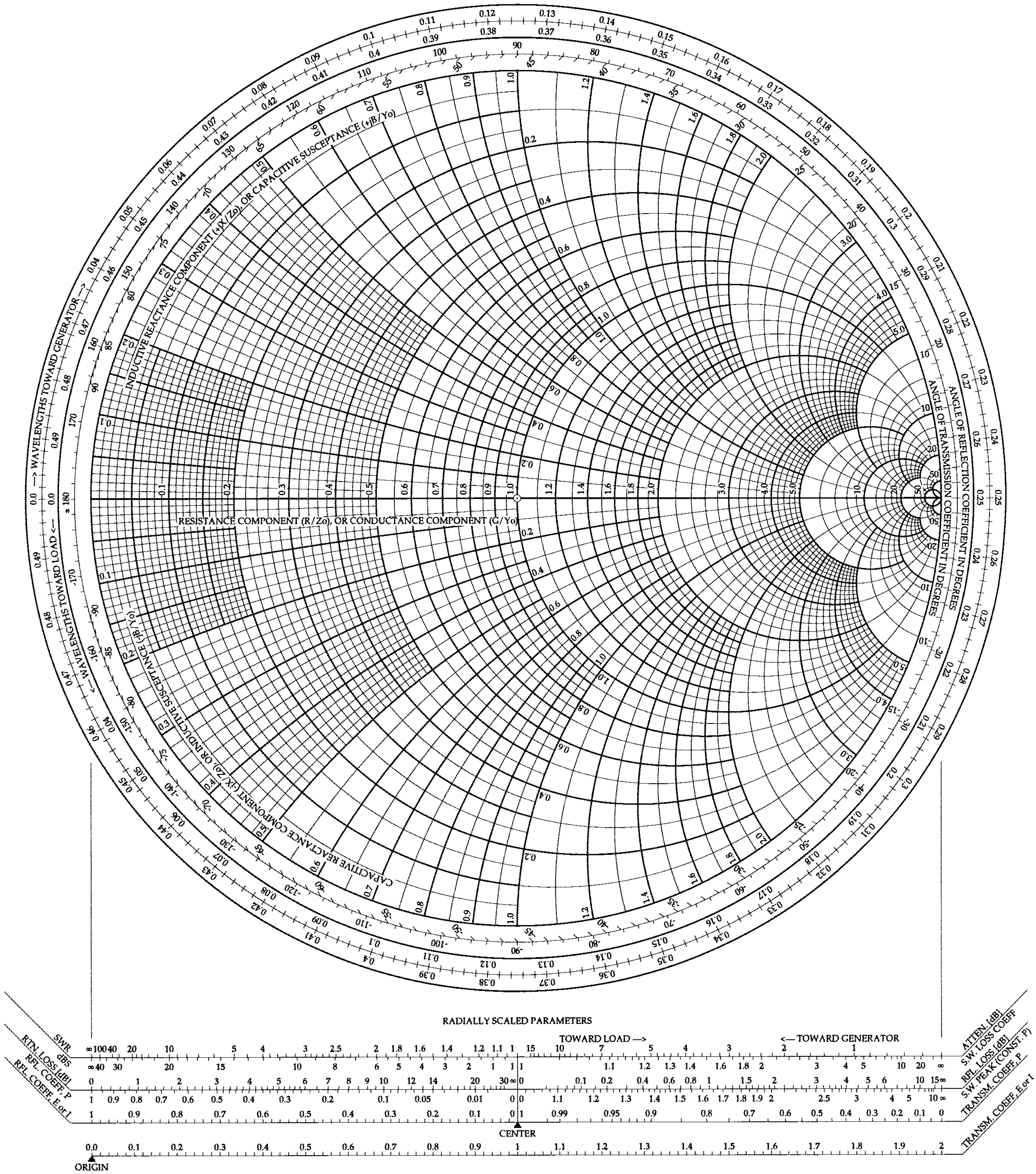
$$\Gamma = \frac{Z - Z_0}{Z + Z_0}$$

return loss = $10\log P_r/P_i = -20\log|\Gamma|$

VSWR = $1+|\Gamma|/1-|\Gamma|$

Candidate Number ....Question Number........

The Complete Smith Chart



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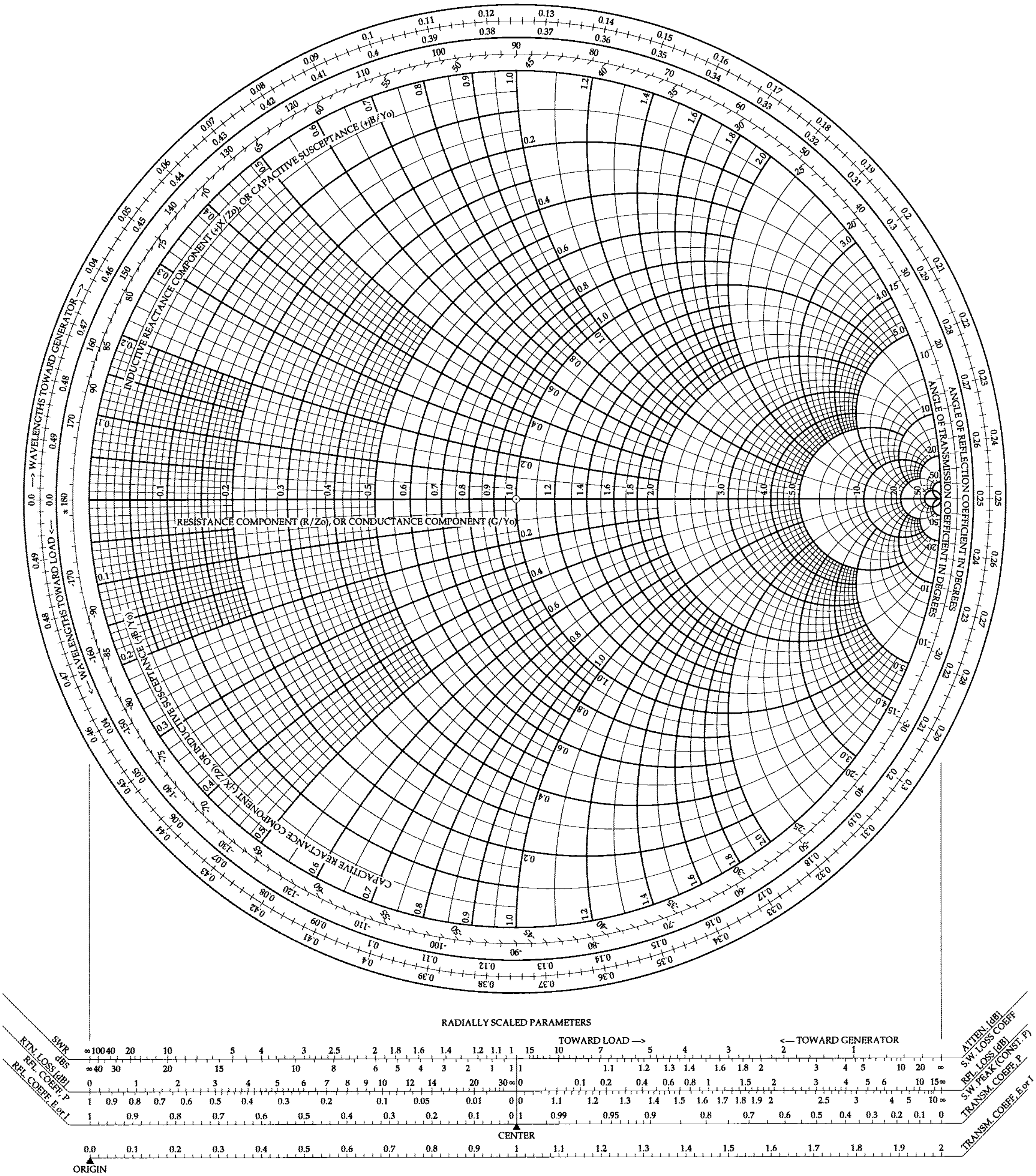

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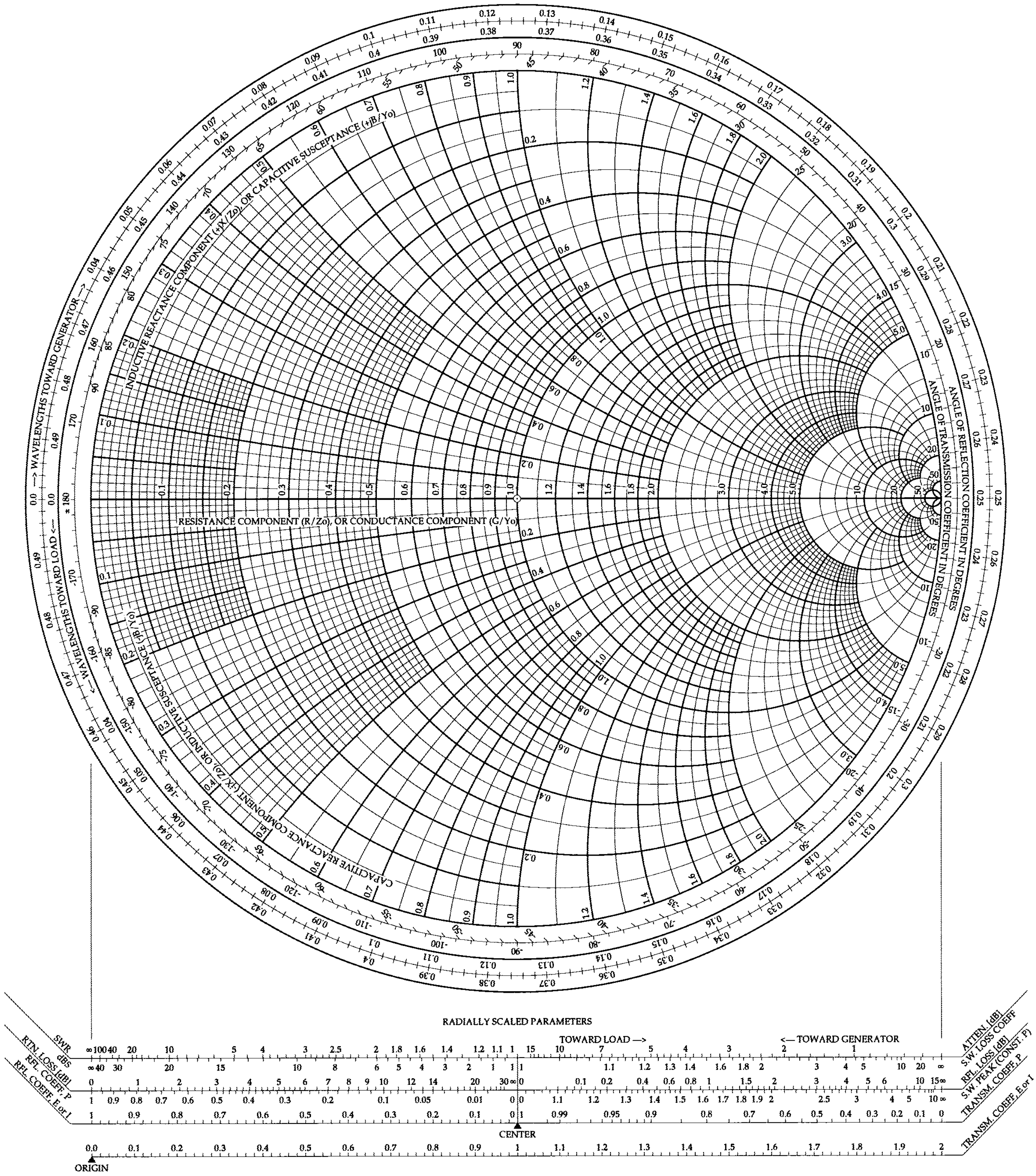

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The Complete Smith Chart